

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) Given the following homogeneous second-order differential equation:

$$4y'' + 8y' + 3y = 0$$

First find the general solution of the given differential equation, then find a particular solution that satisfies the initial conditions $y(0) = 2$, $y'(0) = -5$.

Characteristic equation: $4r^2 + 8r + 3 = 0$
 $(2r+1)(2r+3) = 0$
 $\Rightarrow r_1 = -\frac{1}{2}, r_2 = -\frac{3}{2}$

$$\Rightarrow y(x) = \underline{C_1 e^{-\frac{1}{2}x} + C_2 e^{-\frac{3}{2}x}}$$

$$y'(x) = -\frac{1}{2}C_1 e^{-\frac{1}{2}x} - \frac{3}{2}C_2 e^{-\frac{3}{2}x}$$

$$\text{thus } y(0) = 2 \Rightarrow C_1 + C_2 = 2 \Rightarrow C_1 = 2 - C_2$$

$$y'(0) = -5 \Rightarrow -\frac{1}{2}C_1 - \frac{3}{2}C_2 = -5 \Rightarrow C_2 = 4$$

$$\text{thus } y(x) = -2e^{-\frac{1}{2}x} + 4e^{-\frac{3}{2}x}$$

Problem 2. (4 points) Show that the three functions x , x^2 , e^x are linearly independent in any open interval by checking the Wronskian $W(x, x^2, e^x)$.

$$\begin{aligned} W(x, x^2, e^x) &= \begin{vmatrix} x & x^2 & e^x \\ (x)' & (x^2)' & (e^x)' \\ (x)'' & (x^2)'' & (e^x)'' \end{vmatrix} = \begin{vmatrix} x & x^2 & e^x \\ 1 & 2x & e^x \\ 0 & 2 & e^x \end{vmatrix} \\ &= x \begin{vmatrix} 2x & e^x \\ 2 & e^x \end{vmatrix} - x^2 \begin{vmatrix} 1 & e^x \\ 0 & e^x \end{vmatrix} + e^x \begin{vmatrix} 1 & 2x \\ 0 & 2 \end{vmatrix} \\ &= x e^x (2x - 2) - x^2 e^x + 2e^x \\ &= e^x (2x^2 - 2x - x^2 + 2) \\ &= e^x (x^2 - 2x + 2) = e^x [(x-1)^2 + 1] \geq e^x > 0 \\ &\Rightarrow x, x^2, e^x \text{ are linearly independent.} \end{aligned}$$